

RECAP: Rapid Event-Level Classification of Affected Properties

Final Project Report

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Abstract

This report presents RECAP, an end-to-end system for building-level post-disaster damage assessment using paired pre- and post-disaster satellite imagery. Our approach re-implements the Siamese ResNet-34 baseline from the xBD/xView2 challenge and extends it with polygon-aware feature pooling, difference-based feature representations, and post-hoc temperature scaling to improve confidence calibration. We train and test the model using event-level splits to prevent geographic overlap between disaster scenes. Across the xBD test set, the model performs well on clearly separable classes such as no damage and destroyed, while showing expected difficulty in distinguishing minor from major damage. Calibration improves the reliability of predicted probabilities, allowing confidence scores to better reflect model certainty. We also develop a lightweight Streamlit-based map interface that enables interactive viewing of pre-/post-disaster imagery and model predictions, demonstrating how visual tools can make machine learning outputs more interpretable for decision-makers during disaster response.

1 Introduction

Natural disasters such as hurricanes, earthquakes, and wildfires cause widespread structural damage that must be assessed rapidly to guide emergency response. Traditional field inspections are slow, hazardous, and resource-intensive. Satellite imagery provides broad coverage but is difficult for responders to interpret manually, especially when thousands of buildings must be evaluated. Automated damage classifi-

cation models can accelerate early triage by converting raw imagery into interpretable building-level assessments.

Building damage classification from satellite imagery is challenging for several reasons: paired pre/post images vary in lighting and viewpoint, mid-level damage categories are visually subtle, and models must generalize to new geographic regions and disaster types. Reliability is equally important responders must trust probability outputs to make threshold-based decisions.

RECAP addresses these challenges by:

- Reproducing a Siamese ResNet-34 change-detection baseline using paired satellite tiles.
- Introducing polygon-aware feature pooling, grounding feature extraction within actual building footprints.
- Applying temperature scaling to calibrate confidence scores and improve decision reliability.
- Building a Streamlit visualization interface that overlays predicted damage on a map for intuitive exploration.
- Evaluating performance using event-level splits to ensure realistic generalization.

Together, these contributions create a reproducible, interpretable, and operationally relevant prototype for automatic post-disaster damage assessment. We hypothesize that polygon-aware pooling and difference-based features will improve separation between subtle damage classes, and that temperature scaling will improve confidence reliability without sacrificing accuracy.

2 Related Work

2.1 Building Damage Mapping

The xBD dataset [1] is the largest publicly available benchmark for building level post-disaster damage assessment. It provides paired pre- and post-disaster satellite imagery, polygonal building footprints, and four damage severity labels. xBD has driven much of the recent progress in the field, including the xView2 challenge, which showcased deep-learning approaches capable of mapping structural destruction after earthquakes, hurricanes, wildfires, and floods. Most competitive solutions rely on convolutional neural networks (CNNs) applied to cropped building tiles or segmentation-based pipelines that localize structures before estimating damage. These methods demonstrate that combining pre/post imagery with building-level supervision is essential for capturing spatial change patterns.

2.2 Siamese Networks for Change Detection

Siamese architectures, such as FC-Siam-conv [2], have become standard for modeling image-based change detection. By processing pre- and post-disaster tiles through identical shared-weight encoders, these networks extract comparable features that can be subtracted, compared, or fused to quantify structural changes. This design naturally aligns with damage assessment tasks, where the goal is to distinguish unchanged buildings from those exhibiting partial or complete destruction. Siamese models also reduce parameter count and enforce consistent feature representation across both time points, improving performance when labeled data is limited.

2.3 Probability Calibration

Deep neural networks often exhibit overconfident or miscalibrated probabilities, which can mislead downstream decision-making in high-stakes applications like disaster response. Temperature scaling [3] is a simple and widely adopted post-hoc calibration method that rescales logits to

align predicted confidence with empirical accuracy. Calibration does not change the underlying class predictions but improves probability trustworthiness, enabling responders to apply meaningful thresholds (e.g., identifying buildings that exceed a 70% likelihood of major or destroyed damage). Calibrated models are therefore more interpretable and operationally useful.

3 Dataset

3.1 xView2/xBD Overview

The xBD dataset [1] is one of the largest resources for post-disaster building damage assessment. It provides high-resolution pre- and post-disaster satellite imagery, polygonal building footprints, and four standardized damage categories ranging from no damage to complete destruction. The dataset spans diverse global disaster events, including hurricanes, earthquakes, floods, and wildfires, captured across multiple geographic regions and architectural styles.

With more than 45,000 image chips and over 850,000 annotated building instances, xBD offers sufficient scale for robust supervised learning and evaluation. The pairing of pre- and post-event imagery enables models to learn explicit change representations rather than relying solely on static cues, while the breadth of events encourages generalization beyond individual disaster types. These features make xBD well-suited for building-level damage classification.

3.2 Image Pair Construction

For each annotated building footprint in the xBD dataset, we generate a pair of spatially aligned image tiles to capture structural conditions before and after a disaster. A pre-disaster tile is cropped according to the footprint’s geographic extent, and a corresponding post-disaster tile is extracted from the identical location to ensure pixel-level alignment. Both tiles are resized to 256×256 pixels, normalized using ImageNet statistics, and optionally augmented to improve robustness against illumination changes, color variation, and minor viewpoint differences.

This paired-tile approach is central to RECAP’s change-detection capability. By presenting the model with temporally synchronized imagery, the system learns structural differences that emerge after a disaster, rather than relying on static visual cues that may vary across geographic regions or disaster types.

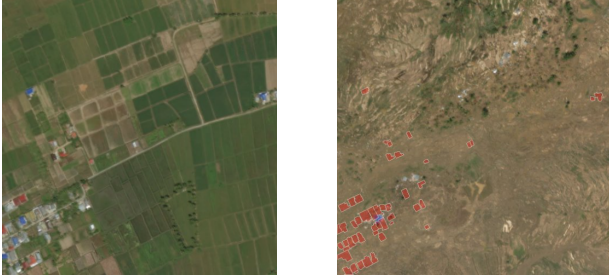


Figure 1: Example pre- and post-disaster tiles. Pre-disaster (left) and post-disaster (right) satellite tiles extracted from the same geographic location using building footprint polygons. Spatially aligned image pairs allow the model to observe structural changes directly and form the basis of effective building-level damage classification.

3.3 Event-Level Train/Validation/Test Splitting

To evaluate the model under realistic deployment conditions, we use an event-level splitting strategy in which entire disaster events, rather than individual building tiles, are assigned to the training, validation, or test sets. This prevents geographic leakage, where visually similar structures from the same region could otherwise appear across multiple splits and artificially inflate performance. Our evaluation therefore more closely mimics real deployments, where models must operate on entirely new disaster events rather than memorized regions.

The composition of the held-out test set is shown in Figure 2. The test distribution shows that most buildings fall under the no-damage category, while minor and major damage categories have substantially fewer examples. This imbalance highlights why event-level splitting is essential: when entire events are held out, the

model must generalize across disaster types, architectural styles, and varying class proportions.

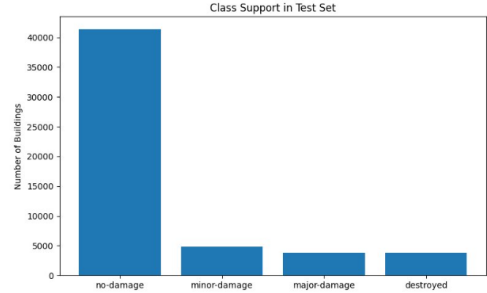


Figure 2: Class distribution in the test set. Distribution of building damage labels in the event-level test set. The prevalence of no-damage structures and the relative scarcity of minor and major damage examples reflect the natural imbalance present in real disaster events.

4 Methodology

4.1 Model Architecture and Inference Pipeline

This work employs a two-stage framework for post-disaster damage assessment using the xBD dataset. The model operates at the building-instance level and integrates (i) a polygon-aware Siamese damage classifier, and (ii) a lightweight building-segmentation backbone for spatial filtering. All computations run offline using local pretrained weights for reproducibility.

Siamese Tile Backbone

We use a ResNet-34 encoder [4] as a shared feature extractor for both pre- and post-disaster satellite tiles. Given an input tile $X \in R^{3 \times H \times W}$, the encoder produces a spatial feature map:

$$F = \text{Encoder}(X) \in R^{C \times H' \times W'},$$

where $C = 512$ for ResNet-34. The encoder is implemented by stripping the final average pooling and fully connected layers from a standard ResNet-34:

$$\text{Encoder} = \text{ResNet34}_{1:-2}.$$

The backbone supports flexible input shapes (e.g., $(3, H, W)$, $(1, 3, H, W)$, $(B, 3, H, W)$), as enforced by a shape-normalization layer prior to the forward pass.

Polygon Rasterization and Feature Pooling

Each building instance is represented by a polygon P defined in original image resolution. To align polygons with the feature map F , they are rescaled by:

$$\tilde{x} = x \cdot \frac{W'}{W}, \quad \tilde{y} = y \cdot \frac{H'}{H},$$

where (W', H') is the spatial resolution of F . A binary mask $M_P \in \{0, 1\}^{H' \times W'}$ is generated by rasterization of the polygon. Building-level feature vectors are obtained via masked global average pooling:

$$v = \frac{\sum_{i,j} F_{:,i,j} M_P(i,j)}{\sum_{i,j} M_P(i,j) + \epsilon}, \quad v \in R^C.$$

This produces one pooled vector for the pre-image (v_{pre}) and one for the post-image (v_{post}). This operation averages features only inside each building footprint, so the resulting vectors focus on the structure itself rather than surrounding background.

Siamese Feature Construction

Following established methods for change detection, we construct a $4 \times$ concatenated representation:

$$z = \begin{bmatrix} v_{\text{pre}} \\ v_{\text{post}} \\ v_{\text{post}} - v_{\text{pre}} \\ |v_{\text{post}} - v_{\text{pre}}| \end{bmatrix} \in R^{4C}.$$

In our case, $4C = 2048$ dimensions. This representation empirically improves separability between subtle change types (e.g., minor vs. major damage).

Damage Classification Head

The classifier is a three-layer multilayer perceptron:

$$\text{MLP}(z) = W_3 \sigma(W_2 \sigma(W_1 z + b_1) + b_2) + b_3,$$

where $\sigma(\cdot)$ denotes ReLU activation. Dropout ($p = 0.3$) is applied between layers to reduce overfitting. The output layer predicts one of the four xBD classes:

$$Y = \{\text{no-damage, minor, major, destroyed}\}.$$

Building Segmentation Module

For completeness, a secondary FCN-ResNet50 segmentation model [5] is included to identify building footprints where polygon annotations may be missing or incomplete. We use a fully offline variant:

$$\text{FCN-ResNet50}(W_{\text{local COCO}}),$$

where pretrained COCO weights are sourced from a locally stored checkpoint. The classifier head is replaced with a one-channel logistic output for binary building segmentation.

4.2 Inference Pipeline

The end-to-end prediction workflow is as follows:

1. Load pre- and post-disaster tiles.
2. Load polygon annotations from xBD JSON files.
3. Rasterize polygons at feature-map resolution.
4. Extract pooled features ($v_{\text{pre}}, v_{\text{post}}$).
5. Construct Siamese feature vector z .
6. Apply the MLP classifier to obtain the predicted damage class.
7. Optionally overlay predicted polygons with color-coded masks:

$$\text{Color}(y) \in \{\text{green, yellow, orange, red}\}.$$

8. Aggregate predictions across tiles to compute accuracy, macro F1, per-class precision/recall, and confusion matrices. Macro F1 computes F1 independently for each class and then averages them, giving rare damage categories equal weight.

Demo and Visualization

The `demo_random_tiles.py` script randomly samples tiles from the test split, performs the full pipeline, and produces predicted overlays, ground truth overlays, global evaluation metrics, per-class metrics, and instance-level error statistics. These figures are exported to `artifacts/demo_overlays/` for presentation use.

5 Calibration

5.1 Motivation

In disaster response, accurate predictions aren't enough responders must also trust the model's confidence scores. Emergency personnel often make threshold-based decisions, such as inspecting any building with more than a 70% predicted likelihood of being major or destroyed. In these scenarios, the reliability of predicted probabilities is as important as the predicted class itself. Overconfident or poorly calibrated probabilities can lead to misallocated resources, delayed inspections, or overlooked structures that appear safe but are not.

Deep neural networks are known to exhibit systematic overconfidence, often assigning high probabilities even when predictions are incorrect [3]. This miscalibration reduces interpretability and undermines operational decision-making, making probability reliability essential for deploying damage-assessment models in real disaster environments.

5.2 Temperature Scaling

Temperature scaling [3] is a simple, post-hoc calibration method that adjusts model logits using

a single learned scalar parameter $T > 0$:

$$p = \text{softmax}(z/T).$$

- When $T > 1$, predictions become less confident, reducing overconfidence.
- When $T < 1$, predictions become sharper, increasing confidence.
- Because scaling is applied after training, class boundaries remain unchanged accuracy does not improve or degrade, only probability reliability changes.

Its effectiveness, simplicity, and model-agnostic nature make temperature scaling one of the most widely adopted calibration techniques.

5.3 Reliability Curve Visualization

To assess calibration quality, we compare predicted confidence with empirical accuracy using reliability curves, shown in Figure 3. An ideally calibrated model follows the diagonal line, where predicted probabilities perfectly match observed frequencies.

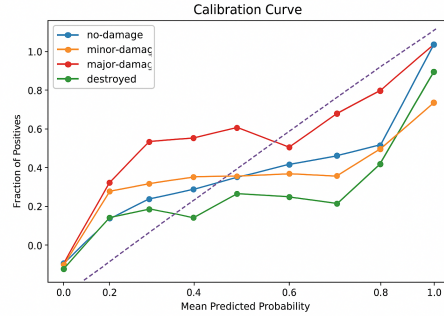


Figure 3: Calibration curves before and after temperature scaling. Reliability curves for each damage class. The uncalibrated model exhibits clear overconfidence, with curves deviating below the diagonal. Temperature scaling brings the curves closer to the ideal reliability line.

Improved calibration does not alter predicted labels but enhances interpretability and decision reliability, particularly for operational users who rely on probability thresholds to prioritize inspections, allocate personnel, and communicate uncertainty to stakeholders.

6 Results

6.1 Quantitative Results

Class imbalance in xBD strongly affects model performance. Most structures fall under the no-damage category, while minor and major damage are underrepresented. This makes it difficult for the model to learn the subtle visual distinctions needed to separate mid-level damage categories.

Model performance reflects this distribution. The confusion matrix in Figure 4 shows strong performance for no-damage and destroyed buildings, which have visually distinct patterns. Minor and major damage categories show considerable overlap, reflecting both their visual ambiguity and limited representation in the dataset.

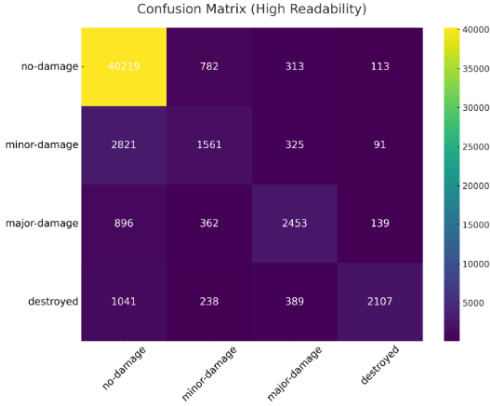


Figure 4: Confusion matrix. Prediction outcomes across the four xBD damage classes. Minor and major damage exhibit the greatest confusion due to subtle structural differences and inconsistent labeling.

Per-class F1-scores in Figure 5 quantify these patterns. Macro F1 gives equal weight to each class, making it appropriate under heavy class imbalance. Extreme categories (no-damage and destroyed) achieve strong performance, while intermediate classes achieve lower scores due to ambiguous structural signatures and insufficient training samples.

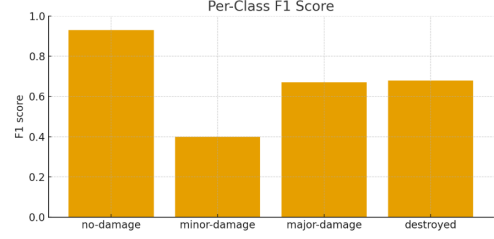


Figure 5: Per-class F1 scores. F1-scores by category, demonstrating strong performance on extreme classes and weaker performance for intermediate damage types.

6.2 ROC Curve Analysis

ROC curves and AUC values measure how well the classifier separates each class from the others in a one-vs-rest setting. Figure 6 presents one-vs-rest ROC curves for each damage class. The model achieves high AUC values for no-damage and destroyed, indicating robust separability. Lower AUC for minor damage reflects the difficulty of detecting subtle structural changes in rare classes.

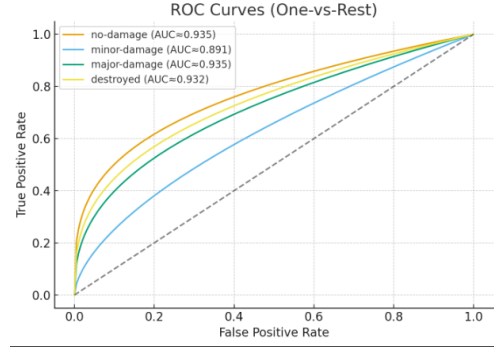


Figure 6: ROC curves (one-vs-rest). Receiver operating characteristic curves for each damage category. High AUC for no-damage and destroyed indicates strong discriminative ability, while lower AUC for minor damage highlights difficulty capturing subtle visual differences.

6.3 Qualitative Examples

Qualitative inspection supports the quantitative findings. Clear structural failures, such as collapsed or missing roofs, are identified with high confidence. Borderline minor vs. major cases ex-

hibit muted visual cues and appropriately lower, calibrated confidence values after temperature scaling.

6.4 Error Analysis

Common sources of misclassification include:

- Subtle differences between minor and major damage
- Occlusion from shadows, vegetation, smoke, or terrain
- Partial roof damage misaligned with polygon annotations
- Small structural changes below satellite resolution

Additional modalities (e.g., multispectral or SAR imagery), improved building footprint annotations, or context-aware models could reduce error rates in intermediate damage categories.

7 Streamlit UI

The Streamlit interface translates RECAP’s predictions into an interactive map for disaster responders. By overlaying color-coded building footprints onto an interactive map, the interface lets users examine model outputs in geographic context. Predicted damage classes are visualized using a standardized color scheme, and hover-based tooltips display calibrated confidence scores, providing both categorical and probabilistic information at the structure level.

The interface uses precomputed predictions for selected disaster events, allowing users to pan, zoom, and inspect regions without computational delays. This supports identification of localized damage patterns, assessment of neighborhood-level impacts, and prioritization of areas requiring further investigation.

The Streamlit UI demonstrates how RECAP’s outputs can be operationalized in decision-support workflows. Its lightweight design, ease of deployment, and clarity of presentation make it an effective prototype for integration into emergency-management systems, research evaluations, and classroom demonstrations.

8 Discussion

Our approach has several strengths. Event-level splits helped the model generalize to completely new disasters, giving a more realistic sense of real-world performance. The model performed well on the no-damage and destroyed classes, where visual differences are clear, and calibration made confidence scores more trustworthy for decision-making. The system is also lightweight and easy to reproduce, making it practical for demos and rapid experimentation. A model that can be quickly deployed and trusted to flag clearly safe versus clearly destroyed buildings directly supports early triage and resource prioritization.

The approach has limitations. Minor vs. major damage remains difficult because the visual differences are subtle and inconsistently labeled. The model also depends heavily on polygon accuracy, and relying only on RGB imagery limits its ability to detect deeper structural changes. Some disaster types have unique visual signatures that do not generalize well across regions.

Several unexpected patterns emerged. Destroyed buildings were easier to classify than damaged ones, and calibration improved user trust without increasing accuracy. Errors often correlated with roof type, materials, shadows, or vegetation. Future work could explore multi-spectral or SAR imagery, transformer-based architectures, better polygon refinement, or uncertainty-aware models to strengthen performance and reliability.

9 Conclusion

We developed RECAP to automate building damage assessment using satellite imagery captured before and after disasters. Through a polygon-aware Siamese encoder, difference-based feature representations, and a calibrated multi-layer perceptron classifier, we extract meaningful building-level damage signals despite highly imbalanced data and subtle structural variations.

A central contribution is the enforcement of event-level train/validation/test splits, which

prevents geographic leakage and ensures evaluation under realistic deployment conditions. Our results show strong performance for the no-damage and destroyed classes, where visual cues are distinct. Performance decreases for minor and major damage, which aligns with known dataset challenges: these categories are visually similar, inconsistently labeled, and underrepresented. This validates our hypothesis that mid-level classes require additional context or richer supervision beyond RGB input.

We also demonstrate the importance of temperature-scaling calibration. Calibration does not increase raw accuracy but improves the reliability and interpretability of predicted probabilities. Calibrated confidence scores provide critical insight into model uncertainty, allowing disaster responders to make threshold-based decisions with a clearer understanding of risk. This is valuable in borderline minor/major cases where misclassifications can lead to inefficient or unsafe field operations.

To make the system usable in practice, we developed a lightweight Streamlit visualization interface. This tool overlays predicted damage levels onto a map with color-coded building polygons and confidence values, offering an intuitive representation of model outputs. Such interfaces highlight how machine-learning predictions can be incorporated into decision-support workflows for emergency management.

RECAP demonstrates that a computationally efficient model can provide actionable building-level assessments when paired with proper data handling, calibrated uncertainty, and user-friendly visualization. Future work may build upon this foundation by incorporating multi-spectral or SAR imagery, adding neighborhood context, leveraging transformer-based architectures, or extending segmentation to refine polygon boundaries. With these enhancements, RECAP could evolve into a robust operational tool for rapid disaster damage assessment and improve situational awareness during infrastructure crises.

References

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